On polarities and computational convexity

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Outline

- Bregman divergences
- Convex duality, mixed parameterizations, and power distance
- Conjugate functions from quadratic polarities and their deformations
- Approximating untractable convex conjugates with neural networks

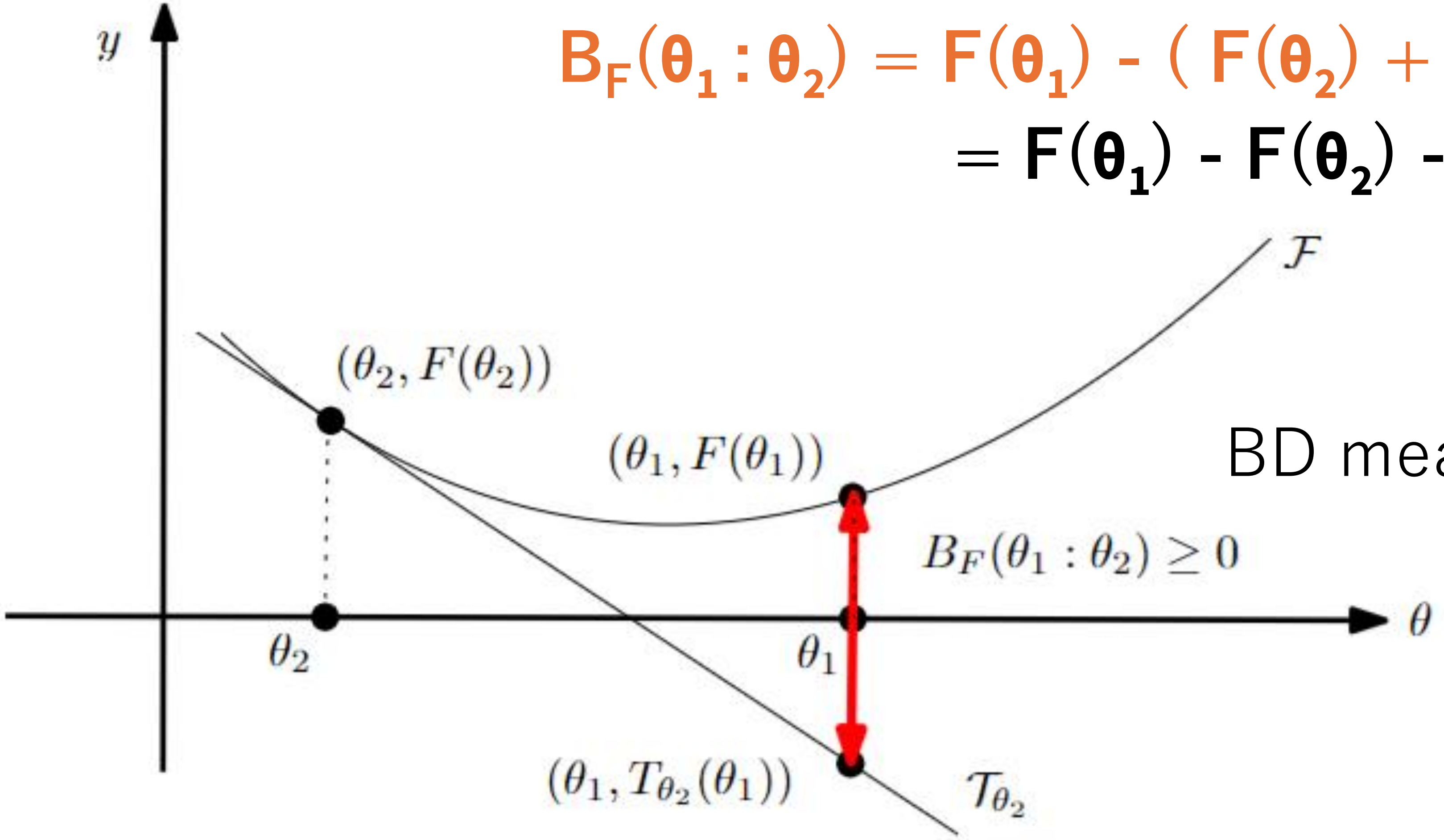
Bregman divergences

- $F: \Theta \subseteq \mathbb{R}^m \to \mathbb{R}$ a strictly convex and smooth real-valued function on a finite-dimensional Hilbert space equipped with inner product <.,.>

Bregman divergence $B_F: \Theta \times \mathrm{Int}(\Theta) \to \mathbb{R}$

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - (F(\theta_2) + \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle)$$

$$= F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle$$



BD measures a convexity gap

Bregman divergences unify various distortions

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - < \theta_1 - \theta_2, \nabla F(\theta_2) >$$

- Quadratic negentropy generator yields ***squared Euclidean divergence:***

$F(\theta) = \frac{1}{2}\, \theta^T Q\, \theta$ with $\Theta = R^d$

$\nabla F(\theta) = Q\theta$

$B_F(\theta_1 : \theta_2) = \frac{1}{2}\, (\theta_2 - \theta_1)^T\, Q\, (\theta_2 - \theta_1) = \frac{1}{2}\, || \theta_2 - \theta_1 ||^2_Q$

Computational geometry L_2^2

- Shannon negentropy generator yields **relative entropy** aka. *Kullback-Leibler divergence :*

$F(\theta) = \Sigma_i\, \theta_i \log(\theta_i)$ with $\Theta = \Delta^d$ (standard/probability simplex)

$\nabla F(\theta) = [\log(\theta_i) - \theta_i]_i$

$B_F(\theta : \theta') = \Sigma_i \{ \theta_i \log(\theta_i / \theta_i') + \theta_i' - \theta_i \} = \Sigma_i\, \theta_i \log(\theta_i / \theta_i')$

$= KL(\theta_i : \theta')$ since $\Sigma_i\, \theta_i = \Sigma_i\, \theta_i' = 1$

Information geometry

Convex duality via Legendre-Fenchel transform

- Legendre-Fenchel transform of a convex function F:

$$F^*(\eta) = \sup_{\theta \in \Theta}\{< \theta, \eta >-F(\theta)\} = \sup_{\theta \in \Theta}\{ H_\theta(\eta) \}$$

Sup of affine functions. Involution $(F^*)^*=F$

F : pointwise supremum of all the affine functions majorized by F

- Convex conjugate: $F^*(\eta) = < (\nabla F)^{-1}(\eta), \eta > - F((\nabla F)^{-1}(\eta))$

  since $\eta = \nabla F(\theta)$. PB: Not in closed form if $(\nabla F)^{-1}$ not available

- Dual Bregman divergences: $B_F(\theta_1 : \theta_2) = B_{F^*}(\eta_2 : \eta_1)$

Dual Fenchel-Young/Bregman divergences

- Young inequality: $F(\theta_1)+F^*(\eta_2) \geq \langle \theta_1, \eta_2 \rangle$
  with equality iff $\eta_2 = \nabla F(\theta_1)$

- Build the Fenchel-Young divergence from this inequality: lhs-rhs ≥0
  $Y_{F,F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle \geq 0$

- Mixed and dual parameterizations of an underlying geometric divergence $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$

$$B_F(\theta_1 : \theta_2) = Y_{F,F^*}(\theta_1 : \eta_2) = Y_{F^*,F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Geometrically, it corresponds to the expression of a point divergence on a Hessian manifold expressed in the θ-chart and the η-chart

Bregman divergences as power distances

$$Y_F(\theta : \eta') := F(\theta) + F^*(\eta') - \langle \theta, \eta' \rangle$$

Use identity $-\langle \theta, \eta' \rangle = \frac{1}{2}\left(\|\theta - \eta'\|^2 - \|\theta\|^2 - \|\eta\|^2\right)$

$$B_F(\theta : \theta') = Y_F(\theta : \eta') = \frac{1}{2}\sigma(\hat{\theta} : \check{\eta}')$$

Power distance $\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$

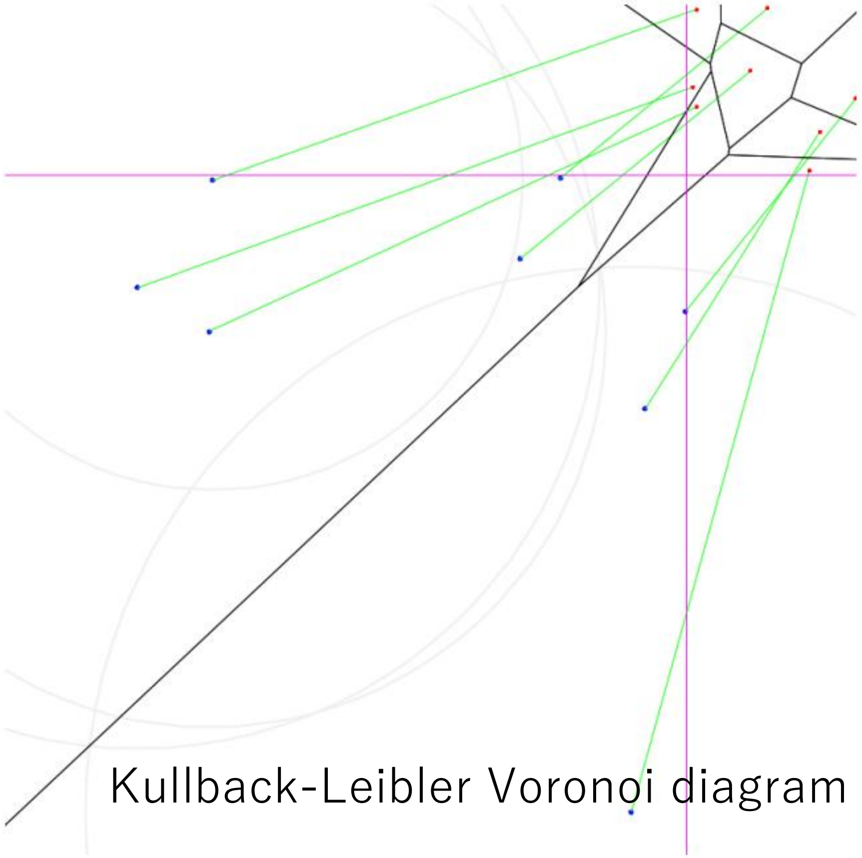
$\hat{p} = (p, w) \quad \hat{p}' = (p', w')$

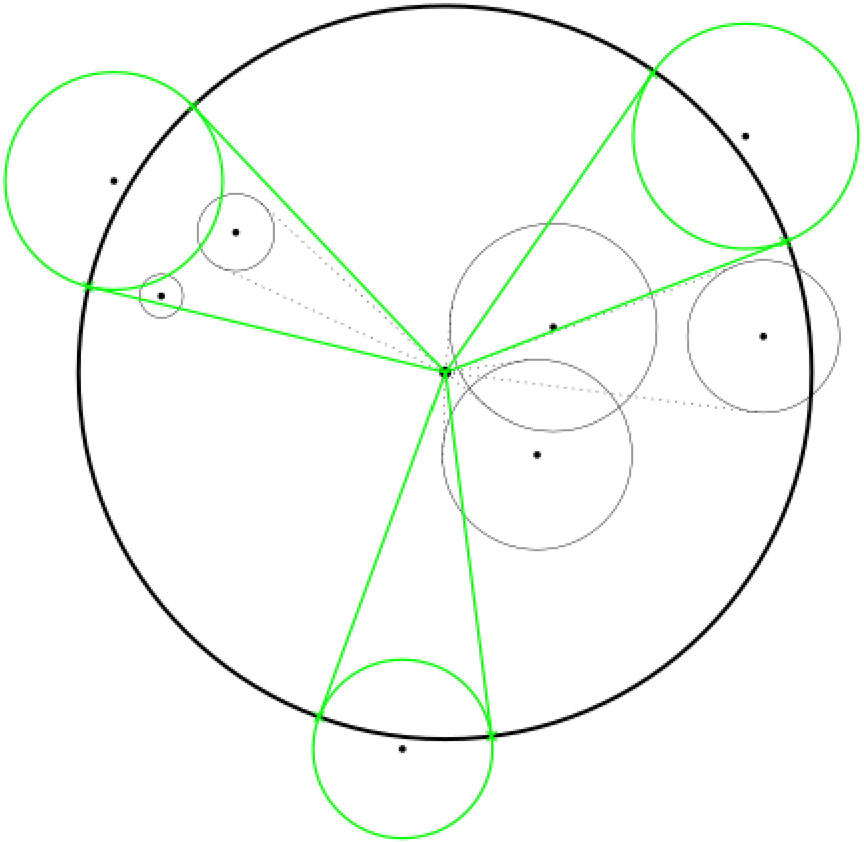
Weighted points expressed in **mixed parameterizations**:

$\hat{\theta} = (\theta, \|\theta\|^2 - 2\,F(\theta)) \qquad \check{\eta} = (\eta, \|\eta\|^2 - 2\,F^*(\eta))$ 2607.24197

Bregman divergences as power distances

Use computational geometry toolbox in information geometry!





Bregman Voronoi in θ as Clipped power diagrams in η

Bregman miniball in θ as Power miniball in η

2607.24197

Convex conjugates from quadratic polarities

Joint work with Basile Plus-Gourdon and Mahito Sugiyama

Polarity of a set A: $\Delta(A) := \{[b] \in \mathbb{R}_{n+1} \quad : \ \forall [a] \in A, \ p(a,b) \geq 0\}$

Quadratic polarity when $p(a,b) := [a]^\top C \, [b]$

Property (Legendre transformation from Legendre polarity). *Let $F : \Theta \to \mathbb{R}$ be a real-valued function and $F^* : H \to \mathbb{R}$ be its Legendre convex conjugate. Then it holds that*

$$\partial \Delta_{\mathcal{L}}(\mathrm{graph}(F)) = \mathrm{graph}(F^*).$$

Legendre polarity $C = C_{\mathcal{L}} := \begin{bmatrix} -I_n & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

n-variate function → Graph (n+1)-dim → Projective space/homogeneous coordinates (n+2)-dim

Legendre transform and gauge transformations

Theorem 1 (Quadratic polarity as transformed convex body of the Legendre polarity). *Let $C \in \mathrm{GL}(n+2)$ and suppose there exists $\mathcal{M}_T \in \mathrm{GL}(n+2)$ such that $C = C_{\mathcal{L}}\mathcal{M}_T^{-1}$, where T is an affine deformation defined by its matrix $\mathcal{M}_T = C^{-1}C_{\mathcal{L}}$ as $T : [b] \mapsto \mathcal{M}_T[b]$. Then for any $A \subset \mathbb{R}^{n+1}$, we have $\Delta_C(A) = T(\Delta_{\mathcal{L}}(A))$.*

Theorem 2 (Quadratic polarity as Legendre polarity on deformed convex body). *Let $C \in \mathrm{GL}(n+2)$ and suppose there exists $\mathcal{M}_S \in \mathrm{GL}(n+2)$ such that: $C = \mathcal{M}_S^{\top}C_{\mathcal{L}}$, where S is an affine deformation defined by its matrix $\mathcal{M}_S = C_{\mathcal{L}}C^{\top}$ as $S : [a] \mapsto \mathcal{M}_S[a]$. Then for any $A \subset \mathbb{R}^{n+1}$: $\Delta_C(A) = \Delta_{\mathcal{L}}(S(A))$.*

$$\mathcal{M}_T = C_{\mathcal{L}}\,\mathcal{M}_S^{-\top}\,C_{\mathcal{L}} \quad \Longleftrightarrow \quad \mathcal{M}_S = C_{\mathcal{L}}\,\mathcal{M}_T^{-\top}\,C_{\mathcal{L}}$$

Fenchel-Young divergences as inner products (GPU)

F a Legendre-type strictly convex function with convex conjugate F*, (F*)*=F
Fenchel-Young divergences are equivalent to Bregman divergences:

$$\begin{aligned} Y_F(\theta_1 : \eta_2) &:= F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle = Y_{F*}(\eta_2 : \theta_1) \geqslant 0 \\ &= \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle \end{aligned}$$

Geometric interpretation: Legendre transform considered as a polarity

$$Y_F(\theta_1 : \eta_2) = (\tilde{\theta}_1^{\uparrow})^\top C_{\mathcal{L}}\, \tilde{\eta}_2^{\uparrow}, \quad C_{\mathcal{L}} := \begin{bmatrix} -I & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \Rightarrow Y_F(\theta_1 : \eta_2) = \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle$$

where $\tilde{\theta}^{\uparrow} := \begin{bmatrix} \theta \\ F(\theta) \\ 1 \end{bmatrix}, \quad \tilde{\eta}^{\uparrow} := \begin{bmatrix} \eta \\ F^*(\eta) \\ 1 \end{bmatrix}$ are homogeneous vectors of points on the graphs of potential functions

Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv: 2606.09077

Convex conjugates via neural networks

Joint work with Basile Plus-Gourdon

2606.09077

- Conjugate $F^*(\eta) = \langle (\nabla F)^{-1}(\eta), \eta \rangle - F((\nabla F)^{-1}(\eta))$

may not be in closed form if $(\nabla F)^{-1}$ not available or computationally intractable.

- Conjugation map: $\theta^*(\eta) = \operatorname*{argmax}_{\theta \in \Theta} \{\langle \theta, \eta \rangle - f(\theta)\} = \nabla f^*(\eta)$

parameterize a *residual network* $\theta_\varphi(\eta) = \eta + \mathrm{MLP}_\varphi(\eta)$, with the multilayer perceptron (MLP)

- Train NN with loss function : $\mathcal{L}(\varphi) = \mathbb{E}_\eta\left[f(\theta_\varphi(\eta)) - \langle \theta_\varphi(\eta), \eta \rangle\right]$
- Once trained, we approximate the convex conjugate by

$$\hat{f}^*(\eta) = \langle \theta_\varphi(\eta), \eta \rangle - f(\theta_\varphi(\eta))$$

- We use gauge freedom to stabilize numerically the NN learning

Summary

- Bregman divergences are coordinatized dissimilarities on Hessian manifolds expressed in dual charts as dual Bregman divergences or in the mixed parameterization as a Fenchel-Young divergence or equivalently power distance in domain of R^n.

  → Computational information geometry

- Conjugate functions defined via quadratic polarity allows one to express the divergence as inner products (fast on GPU) and consider affine deformations

- When not available in closed form, conjugates can be approximated using neural networks where the gauge freedom is used to stabilize the learning/accuracy.

  → Neural information geometry

References

- Quadratic polarity and polar Fenchel-Young divergences from the canonical Legendre polarity, arXiv:2603.04812
- Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv:2606.09077
- Minimum enclosing Bregman balls made easy, arXiv:2607.24197

Preconditioning Neural Legendre transform

paraboloid $Q(\theta) = \frac{1}{2}\|\theta\|^2$ is its own Legendre conjugate: $Q^* = Q$

with conjugate map identity: $\theta^*(\eta) = \eta$ (initialization in NN)

Taylor approximation:

$$f(\theta) \approx f(\theta_0) + \frac{1}{2}(\theta - \theta_0)^\top H(\theta - \theta_0), \quad H = \nabla^2 f(\theta_0).$$

Under the change of variable $z = H^{1/2}(\theta - \theta_0)$ this becomes $f \approx f(\theta_0) + \frac{1}{2}\|z\|^2$

learning task in z-space is approximately that of the canonical paraboloid

$$(f')^{*\prime}(\eta') = H^{-1/2}\,\theta^*(H^{1/2}\eta') - \theta_0 \;=:\; z^*(\eta')$$

$$f'(z) = f(H^{-1/2}z + \theta_0) - f(\theta_0),$$

$\eta' = \eta H^{-1/2}$ (row-vector convention) $\quad \theta^*(\eta) = H^{-1/2} z^*(\eta H^{-1/2}) + \theta_0$